

Why we still struggle to stay connected

Even the most optimistic coverage reports admit that vast stretches of rural India, particularly hilly regions, forests, and farmlands, remain underserved. Towers exist, but capacity is limited, and fibre backhaul often stops far short of remote habitations.

Consider this: a farmer in Ladakh walks several hundred metres each morning just to find a signal strong enough to call his supplier. In certain border villages, people still queue at the one spot where a satellite phone catches reception. This 'last-mile challenge' is not purely a rural issue. Connectivity blackspots exist on highways, railways, air routes, and even industrial sites.

As India's infrastructure modernises, so does its connectivity demands. High-speed trains such as the Vande Bharat Express run at 160-200km/hr, speeds at which traditional mobile handovers become tricky. Aircraft require reliable in-flight communication; ships crossing oceans, need stable data links; factories and mines demand secure IoT networks spread over vast or shifting terrain.

These use cases expose the limits of conventional terrestrial networks. Even with dense tower deployment, fibre-optic backhaul, and advanced

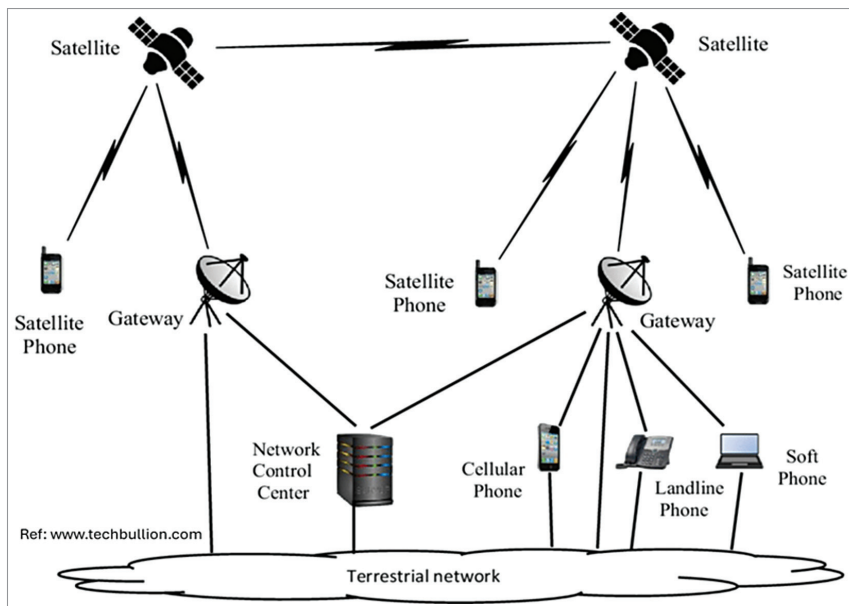


Fig. 1: SATCOM and other networks integration (Credit: www.techbullion.com)

5G architectures, physical and economic boundaries persist: and that is where the sky comes in.

The rise of the sky network

Satellite communication is not new; it has quietly powered television broadcasts, GPS, defence communication, and weather monitoring for decades. What is new is its reinvention for the modern data era. Traditionally, satellites orbiting high above Earth (around 36,000km in geostationary orbit) provided broad coverage but suffered from high latency. Sending a signal to space and back could take half a second, acceptable for television but not for

real-time internet.

Then came the era of low earth orbit (LEO) constellations, satellites placed just a few hundred to a couple of thousand kilometres above the surface. By slashing latency to 20-40 milliseconds, they made broadband from space feasible. Companies like SpaceX's Starlink, OneWeb, and Amazon Kuiper have launched thousands of LEO satellites, creating constellations that blanket the globe. These systems deliver broadband speeds comparable to terrestrial networks, even across deserts, oceans, and mountains.

Meanwhile, ISRO and several Indian startups are developing indigenous small-satellite constellations, ensuring the nation is not left behind in this new communications race. As launch costs drop, thanks to reusable rockets and miniaturised satellites, space-based connectivity is becoming economically viable.

5G and SATCOM: A match made in technology heaven

While satellites extend reach, 5G brings intelligence, speed, and flexibility. For the first time in telecom history, 5G has been designed around use cases, smart cities,

India's quiet SATCOM revolution

While much of the public conversation revolves around fibre and 5G towers, satellite communication is already embedded in India's digital fabric.

- Over 125,000 ATMs rely on satellite links to securely process transactions in remote regions.
- Around 50,000 petrol pumps use SATCOM for payment authorisation and monitoring.
- Disaster management agencies, defence units, and remote schools use VSAT (Very Small Aperture Terminal) networks daily.

Now, with the integration of 5G, this foundation is expanding. Telecom operators such as Airtel and Jio are exploring partnerships with satellite firms, while startups are developing indigenous satellite broadband systems aligned with ISRO's vision of an Indian LEO constellation.

As 3GPP's Release 19 (currently in publication) introduces more specialised provisions for non-terrestrial networks, India's policy ecosystem is preparing for commercial deployment.